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PI Name:	Fernandez-Pello, Carlos Ph.D.		
Project Title:	Forced Flow Ignition and Flame Spread Test (FIST)		
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Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
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Zip Code:	94720-1740	**Congressional District:**	9
Comments:			
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No. of PhD Candidates:	2	**No. of Master' Degrees:**	
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No. of Bachelor's Candidates:	6	**Monitoring Center:**	NASA GRC
Contact Monitor:		**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: Received NCE to 9/30/2008 per PI (8/08)		
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	The Forced flow Ignition and Flame Spread Test (FIST) aims to study the effects of gravity on the flammability diagrams of solid combustible materials, and the fire properties derived from them. The final objective of the research is to develop a new flammability apparatus that would better reflect the potential ambient conditions of space-based environments, i.e., microgravity, low velocity flow, oxygen concentrations below and above that of air, reduced pressure, and use it to obtain more appropriate flammability diagrams for practical materials (plastics, cable jackets, electronic circuit boards, etc.) that will be found in space-based facilities. The strong possibility of an accidental fire occurring in a space-based facility during a long mission is a primary safety concern. Consequently, it is of critical importance to characterize the fire properties of materials that are used. FIST is similar to the ASTM E-1321 LIFT apparatus, except the fuel sample is mounted in small-scale wind tunnel so that a low-velocity oxidizer flow of specified oxygen concentration can be forced over the sample surface during the		

Task Description:	test. The FIST protocol includes both ignition and flame spread testing. For an ignition test, a 30 mm by 30 mm test specimen is embedded flush in the substrate of the wind tunnel so that the gas flow boundary layer generated over its surface is not disrupted. Two radiant heaters are used to irradiate the sample with an approximately uniform heat flux of known intensity. Ignition is induced at fuel sample's trailing edge with an electrically heated wire coil that acts as a pilot. Measured quantities for a particular material include the time to ignition as a function of ambient oxygen concentration and flow velocity, as well as the critical (minimum) mass and heat flux for ignition. Alternatively, flame spread tests can be conducted using larger samples (100 mm long by 30 mm wide). The sample is exposed to a heat flux that decays in the longitudinal direction. The sample is ignited locally at one end, and a flame spreads over teh sample surface. The rate of flame spread is measured by digitizing infrared images and taking the slope of the flame position vs. time. The ignition and flame spread data are plotted on a flammability diagram which is a measure of a material's overall flammability. To complement the experimental program, theoretical modeling of the ignition, burning, and flame spread process is performed. This modeling includes a description of the solid-phase heating and fuel generation process through a customized pyrolysis model. The pyrolysis model is coupled to the CFD-based Fire Dynamics Simulator for calculation of gas-phase phenomena (fluid mechanics, combustion, heat transfer, radiation, etc.). The combined model is used to calculate ignition, burning, and flame spread processes for different materials, environmental conditions, and geometries. One of the principal purposes of the modeling effort is to extend via modeling the experimental FIST results to other conditions that have not been tested, i.e. different O2, velocity, or gravity levels.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Composites are finding increased use in the transportation and construction industries. One barrier to their widespread adoption is the increased flammability of polymer based composites when compared with other materials. In the process of developing a test methodolgy for flammability of composites intended for use in space structures, the FIST project is developing a better understanding of composite materials' flammability useful to ground based design and manufacture.
	Flame spread modeling work during FY2007 focused on simulation of the KC-135 flame spread experiments conducted in October 2004. At lower heat flux levels, the modeled flame spread rate is underpredicted but matches the experimental data reasonably well (within 35%). However, at higher heat flux levels, the experimental flame spread rate is overpredicted by almost a factor of 2. This overprediction may be partially due to a discrepancy between the experimental preheat procedure and its representation in the simulations. It is anticipated that a multi-step gas-phase combustion mechanism and incorporation of the effect of ambient oxygen concentration into the condensed-phase decomposition model could improve model predictions. While inclusion of a comprehensive gas-phase reaction mechanism into a flame spread model is not feasible, a three or four-step mechanism that could account for (at least approximately) the effect of ambient oxygen concentration on the combustion process would be quite useful. Similarly, a simplified model that can account for the effect of ambient oxygen concentration on the thermo-oxidative decomposition of solid fuels could also potentially improve model predictions. During FY2007, the focus of the pyrolysis modeling was to account for the effect of oxygen on the thermo-oxidative decomposition of combustible solids. The pyrolysis model was extended to solve the gas-phase energy equation (inside the decomposing solid, not in the "ambient" gas-phase outside of the decomposing solid because that is handled by a CFD code). In previous versions of the model, thermal equilibrium was assumed between the solid and gas-phase, making it unnecessary to solve the gas-phase energy equation. However, by solving the gas-phase energy equation, it becomes possible to consider thermal disequilibrium between the gas-phase and the condensed-phase inside the decomposing solid. More importantly, the heat released due to the gas-phase combustion of pyrolysate inside the decomposing can now be included in simulations because the model was also extended to include a user-specified number of homogeneous gas-phase reactions. This addition has made it possible to simulate the effect of ambient oxygen concentration on the thermo-oxidative decomposition of solids. Efforts have been made to extend the pyrolysis model to "practical" materials. Preliminary modeling of a laminated polyester composite has been conducted, with encouraging results. In the FY2006 report, it was stated that an analytical model for opposed-flow flame spread rates was under development. This model predicts that flame spread is a balance between the heat-transfer to the solid ahead of the pyrolysis front and finite-rate chemical kinetics. This model was applied to existing, well-known flame spread data as well as the flame spread data gathered on the KC-135 with excellent agreement. In order to fully understand the predictive capabilities of this analytical model, additional flame spread experiments at high opposed-flow velocity subject to the heat flux levels used in the microgravity experiments were needed. Flame spread experiments in FY2007 were focused on obtaining this data. To gain some estimate of the experimental error, each test was performed a total of 3 times. The average percent variation from the mean was 21% for PMMA and 14.2% for PPG 30%. Clear trends in the experimental data were seen and matched by the analytical model. As the external heat flux level is increased, the flame spread rate increases for the same opposed-flow velocity. At lower heat fluxes, the spread rate behaves much as it does in the case with no heat flux. For a range of opposed-flow velocities, the spread rates are insensitive to changes in opposed-flow velocity due to the dominance of buoyancy-induced flow. The maximum flame spread rate occurs in this buoyancy-driven range and is therefore not seen experimentally in normal gravity. However, it does seem as if increasing the heat flux level has the same results as increasing the ambient oxygen concentration – the maximum spread rate shifts to the right, occurring at higher and higher opposed-flow velocities. Not only did the analytical model qualitatively match the trends in the experimental data for a wide range of test conditions, it was seen that the average difference between the model and the data over both materials and all conditions is 20%. During FY2007, a new focus area for the FIST project was introduced: the ignition delay of polymeric materials in exploration atmospheres. Because future space vehicles will operate with a total cabin pressure of 8.5 psia with an oxygen concentration of 32%, it is crucial to understand fire behavior in this atmosphere. To accomplish this, a pressure chamber with the ability to control the ambient oxygen concentration will be used. The CIR chamber, previously used on the KC-135, was transported from Lawrence Berkeley Labs to the UC Berkeley campus with this goal in mind. In order to perform ignition delay tests in "exploration atmospheres," the FIST tunnel was modified so that it fit inside the chamber and a frame was built to support the tunnel within the chamber. A heater mounting system was manufactured, along with a sample holder. Feedthroughs were obtained to connect the fan power, igniter power, heater power, and

thermocouples to the power supplies and data acquisition equipment located outside of the chamber. In order to sense and control the pressure in the chamber, an appropriate pressure transducer and vacuum pump were obtained. The chamber was also connected to an oxygen supply to control the concentration inside. A system for the extinguishment of the flame has also been implemented. Ignition delay tests in regular atmospheric conditions have begun to serve as a basis of comparison for later tests. Once this series of baseline tests are completed, tests will be done in low pressure/normal oxygen, normal pressure/high oxygen, and low pressure/high oxygen environments with both PMMA and PPG 30%. To date, ignition delay tests in standard air have been performed with PMMA for heat flux levels of 9 to 12 kW/m2 with airflow velocities of 0.4, 0.7, and 1 m/s.

In FY2006, it was decided that the FIST apparatus should be modified to gain the ability to perform heat release rate measurements. Before heat release rate measurements can be performed, some modeling was done to understand how to setup the FIST tunnel to obtain the best accuracy and to determine the appropriate additional equipment needed. The focus was on two main aspects: how and where to sample the combustion gases. The measurement precision can vary whether we measure only oxygen, both oxygen and carbon dioxide, or oxygen, carbon dioxide and carbon monoxide. Because the carbon monoxide generation rate is very low for PMMA, the third method does not really improve the accuracy; results found with the first two methods are similar. For simplicity, the first experiments will focus on only oxygen measurement. An initial theoretical calculation was performed to determine the expected oxygen depletion in the outflow during combustion. Using equations of reaction, combustion, and the duct characteristics, we found that the oxygen will decrease from 21 to 20.07 percent in volume which, using the oxygen consumption method, corresponds to a heat release rate of approximately 1,100 kW/m². Based on these results, a short calculation shows that an error of ± 100 ppm in the oxygen concentration measurement results in an acceptable error in the heat release rate value of less than 2% error. An oxygen sensor with this level of accuracy was recently obtained. However, in order to obtain the best measurement accuracy, it is necessary for the combustion gases be thoroughly mixed. In order to determine the necessary modifications to the FIST tunnel, a model of the duct was created using FDS and Smokeview. Different duct designs were modeled to determine which was the most effective at mixing the exhaust gases and it was found that the least expensive and easiest to implement is to simply increase the duct length by about 30 cm.

Task Progress:

SUPPLEMENTAL REPORTING FOR FINAL TASK BOOK REPORT

Continuing work begun at the end of FY2007, the ignition delay of polymeric materials in exploration atmospheres was explored. Because future space vehicles will operate with a total cabin pressure of 52.7 – 58.6 kPa with an oxygen concentration of 30 – 34%, it is crucial to understand fire behavior in this atmosphere. To accomplish this, a pressure chamber with the ability to control the ambient oxygen concentration was used. In these tests, polymethylmethacylate (PMMA) was exposed to a range of oxidizer flow velocities from 40 to 100 cm/s and externally applied heat fluxes from 8 to 14 kW/m2. Tests were conducted for a baseline case of normal pressure and oxygen concentration, low pressure (58.6 kPa) with normal oxygen (21%), and low pressure with 32% oxygen concentration conditions to determine the individual effect of pressure and the combined effect of pressure and oxygen concentration on the ignition delay.

Overall, it was found that reducing the pressure to 58.6 kPa while keeping the oxygen concentration at 21% reduced the ignition time by 17% on average. Increasing the oxygen concentration to 32% at 58.6 kPa reduced the ignition time by an additional 10%. The critical heat flux for ignition was also found to decrease in exploration atmospheres. These results show that tests conducted in standard atmospheric conditions will underpredict the ignition of materials intended for use on spacecraft and that, at these conditions, materials are more susceptible to ignition than in current spacecraft atmospheres. These results seemed counterintuitive at first, but they can be explained by considering the effect of pressure on the convective heat transfer coefficient and the mass loss required to reach the lean flammability limit. For forced flow over a flat plate, the convective heat transfer coefficient is proportional to the square root of the pressure. Pressure also affects the required mass loss to reach the lean flammability limit. Ignition occurs when the fuel and air mixture near the igniter reaches the lean flammability limit. As the pressure is reduced, the amount of air is reduced, so less fuel is required for the mixture to be at the lean flammability limit. Because the pyrolysis process has an Arrhenius dependence on temperature, ignition can occur at a lower temperature. When the oxygen is increased, the surface temperature at ignition is even lower because an elevated oxygen concentration can increase the rate of pyrolysis at a given temperature.

In order to more fully understand the effect of pressure on ignition delay, and perhaps find a less hazardous alternative to the proposed space exploration environment, another three series of tests were performed. In all three sets of experiments, the forced flow was fixed at 30 cm/s and the heat flux fixed at 14 kW/m2. In the first set, the pressure was lowered from atmospheric (101.3 kPa) until no ignition occurred with air (21% O2) as the oxidizer. As the pressure is decreased below atmospheric, the ignition delay time decreases. The delay time continues to decrease until about 28 kPa (4 psi) where a minimum occurs. The minimum observed ignition time is 22% lower than the ignition time in sea-level air. If the pressure is decreased below 28 kPa, the ignition time sharply increases until no ignition is seen at a pressure of 7 kPa (1 psi). By looking at the surface temperature traces, three regimes were identified. As the pressure is reduced from atmospheric to about 50 kPa, both the reduction in heat transfer coefficient and in the mass loss required to reach the lean flammability limit are dominant and the ignition time decreases rapidly. Below 50 kPa, the surface temperature at ignition begins to increase, implying that the chemistry time is becoming noticeable. Between pressures of 50 kPa and 27.6 kPa, the reduction in the heat transfer coefficient is competing with the increase in the chemical time so the ignition time still decreases as the pressure is reduced but not as quickly. Below 34.5 kPa, the heat transfer coefficient remains nearly constant as the pressure is decreased. It follows then that the increase in the chemical time is the sole mechanism controlling the ignition time below 27.6 kPa, the pressure where the minimum ignition time occurred.

In order to better understand the combined effect of reduced pressure and elevated oxygen concentration, the second series of experiments were performed in pressures from atmospheric down until no ignition occurred with the oxidizer of air enriched to 30% oxygen. The same trends seen with air as the oxidizer are also seen with oxygen-enriched air. The ignition delay time decreases, reaches a minimum, then begins to increase until no ignition occurs. However, when the oxygen concentration increases the pressures at which the minimum and no ignition occurs are different. The minimum ignition time occurred at a higher pressure of 41 kPa and the no ignition point occurred at a lower pressure of about 3.4 kPa. An elevated oxygen concentration changes the controlling mechanisms. Increasing the oxygen concentration may increase the pyrolysis rate at a given temperature, but it also increases the mass of fuel required to reach the lean flammability limit. The latter effect of oxygen may be responsible for shifting the pressure at which the minimum ignition time occurs. By looking at the surface temperature histories, it is estimated that the influence of oxygen on the pyrolysis rate is strong only at pressures above 70 kPa. At lower pressures, the collision frequency is perhaps small enough that this effect is masked by the increase in the required mass loss to reach the lean flammability limit. The

	gas-phase reaction rate is increased with the increase in oxygen so it is expected that the increase in the chemical induction time will occur at a lower pressure than in air. The change in heat transfer coefficient with pressure is not influenced by the oxygen concentration, so the range of pressures where this mechanism should be considered does not change. In order to fully understand these controlling mechanisms, however, it is necessary to examine the mass loss rate at ignition and is a subject of future research. The third series of experiments were aimed at understanding ignition delay in low-pressure environments where humans could live. In this series, the pressure was reduced while keeping a constant partial pressure of oxygen of 21.3 kPa, in other words, “normoxic” equivalent environments. The ignition delay time is decreased when the oxygen concentration is increased from air at 21% to 30%. However, if the oxygen is increased further no additional decrease in the ignition time is seen. In fact, the normoxic curve follows the curve for 30% oxygen for all but the atmospheric pressure tests. For pressures below atmospheric, the normoxic oxygen concentration varies from 24 to 77%, indicating that any increase in oxygen concentration beyond 24% does not further reduce the ignition delay time because the chemical kinetics are fast, regardless of pressure. Theoretical work in FY2008 focused on integrating the pyrolysis model with FDS 5.2.3. The coupled code, a user’s guide, and a technical reference can be downloaded here: http://code.google.com/ Since many materials and practical systems exhibit two-dimensional behavior while undergoing pyrolysis, the solid-phase model has been extended to two dimensions. The code now includes an option to solve fully two-dimensional conservation equations for species, energy, mass, and momentum. This also makes it possible to account for the effect of buoyancy on pyrolysis and smoldering. Thus, the code can now explicitly account for the effect of gravity level on smoldering and pyrolysis.
Bibliography Type:	Description: (Last Updated: 02/15/2026)
Abstracts for Journals and Proceedings	Lautenberger C, Fernandez-Pello C. "Generalized Pyrolysis Model for Combustible Solids." Annual Fire Conference at the National Institute of Standards and Technology, Gaithersburg, MD, April 4-5, 2007. Annual Fire Conference at the National Institute of Standards and Technology, April 2007. , Apr-2007
Articles in Peer-reviewed Journals	McAllister S, Fernandez-Pello C, Urban D, Ruff G. "Piloted ignition delay of PMMA in space exploration atmospheres." Proceedings of the Combustion Institute, 2009 Jan;32(2):2453-9. http://dx.doi.org/10.1016/j.proci.2008.05.076 , Jan-2009
Articles in Peer-reviewed Journals	Lautenberger C, Fernandez-Pello C. "Generalized pyrolysis model for combustible solids." Fire Safety Journal. 2009 Aug;44(6):819-39. http://dx.doi.org/10.1016/j.firesaf.2009.03.011 (originally reported in August 2009 as "In Press, Corrected Proof, Available online 20 May 2009.") , Aug-2009
Books/Book Chapters	Lautenberger C, Fernandez-Pello C. "Pyrolysis Modeling, Thermal Decomposition, and Transport Processes in Combustible Solids." in "Transport Phenomena in Fires." Ed. M. Faghri, B. Sunden. Southampton : WIT Press/Computational Mechanics, 2008. (Developments in Heat Transfer, Vol 20) p. 209-255., Oct-2008
Dissertations and Theses	Rich D. "Development of the Forced Ignition and Flame Spread Test." Dissertation, University of California, Berkeley, December 2006. , Dec-2006
Dissertations and Theses	Lautenberger C. "Generalized Pyrolysis Model for Combustible Solids." Dissertation, University of California, Berkeley, December 2007. , Dec-2007
Papers from Meeting Proceedings	Lautenberger C, Fernandez-Pello C. "A Generalized Pyrolysis Model for Simulating the Gasification of Noncharring, Charring, Intumescent, and Smoldering Combustibles." 5th International Seminar on Fire and Explosion Hazards, Edinburgh, Scotland, UK, April 23-27, 2007. 5th International Seminar on Fire and Explosion Hazards, Edinburgh, Scotland, UK, April 23-27, 2007. , Apr-2007
Papers from Meeting Proceedings	McAllister S, Rich D, Lautenberger C, Fernandez-Pello C, Yuan ZG. "Modeling Microgravity and Normal Gravity Flame Spread Rates over Polymers and Polymer/Glass Composites." 5th International Seminar on Fire and Explosion Hazards, Edinburgh, Scotland, UK, April 23-27, 2007. 5th International Seminar on Fire and Explosion Hazards, Edinburgh, Scotland, UK, April 23-27, 2007. , Apr-2007
Papers from Meeting Proceedings	McAllister S, Rich D, Lautenberger C, Fernandez-Pello C, Yuan ZG. "An Analytical Model for Opposed-Flow Flame Spread Rates over Polymers and Polymer/Glass Composites." 5th US Combustion Meeting, San Diego, CA, March 26-28 2007. 5th US Combustion Meeting, San Diego, CA, March 26-28 2007. , Mar-2007
Papers from Meeting Proceedings	McAllister S, Rich D, Lautenberger C, Fernandez-Pello C, Yuan ZG. "Modeling Microgravity and Normal Gravity Flame Spread Rates over Polymer/Glass Composites." 45th AIAA Aerospace Sciences Meeting and Exhibit. Reno, NV, January 8-11 2007. AIAA paper AIAA-2007-740, 2007. , Jan-2007
Papers from Meeting Proceedings	Lautenberger C, McAllister S, Rich D, Fernandez-Pello C. "Effect of Environmental Variables on Flame Spread Rates in Microgravity." 45th AIAA Aerospace Sciences Meeting and Exhibit, Reno NV, January 8-11 2007. AIAA paper AIAA-2007-383 , 2007. , Jan-2007
Papers from Meeting Proceedings	McAllister S, Fernandez-Pello C, Urban D, Ruff G. "Ignition Delay of Combustible Materials in Space Exploration Atmospheres." 38th International Conference on Environmental Systems, San Francisco, CA, June 2008. SAE paper 2008-01-2088. Warrendale, PA : SAE International, 2008. , Jun-2008
Papers from Meeting Proceedings	McAllister S, Lai J, Scott S, Ramirez-Correa A, Fernandez-Pello C, Urban D, Ruff G. "The Effect of Pressure on Piloted Ignition Delay of PMMA." 46th AIAA Aerospace Science Meeting and Exhibit, Reno, NV, January 7-9, 2008. AIAA paper AIAA-2008-1413, American Institute of Aeronautics and Astronautics, 2008. , Jan-2008
Papers from Meeting Proceedings	McAllister S, Fernandez-Pello C, Urban D, Ruff G. "Piloted Ignition Delay of PMMA in Space Exploration Atmospheres." 32nd International Symposium on Combustion, Montreal, Canada, August 2008. Combustion Institute paper PROCI-D-08-00249, Proceedings of the Combustion Institute, 2008. , Aug-2008

Significant Media Coverage	McAllister S, Fernandez-Pello C. "Interviewed for the show, The Universe: Deep Space Disasters, regarding the potential fire risk onboard spacecraft." The Universe: Deep Space Disasters, aired on the Discovery Channel on November 11, 2008., Nov-2008